

Standard Cost and Purchasing Performance Definitions

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Summary

After a difficult start in 2011, we need to harmonize and synchronize the content and the principle of our purchasing performance control system and tools.

For each BU, the performance will be now based on 3 main indicators:

- the PPV (purchase price variance)
- the SIM values evolution
- IMPACT plan dashboards

These 3 indicators, if well managed by MLM, are sufficient to support and to alert COO and CEO.

Specific corrective actions have to be taken if the figures are not in line with the expectations in cost reduction program and if deviations in material cost appear.

Standard Cost

The standard cost in BaaN is based on the purchase price from the supplier (MAT) to which is applied 2 factors, the Freight In Rate (FIR) and the Material Burden Rate (MBR).

For manufactured items, the standard cost additionally includes the sub-contracting cost (OPC, including labour & material), the standard labour hours at direct labour rate (ALB) and the standard labour hours with burden rate (AOH).

1. Manufacturing components

The standard cost of an item is calculated once a year. The purpose of the standard cost is to use it as a reference to measure the variances during the year (it is also the basis for BaaN to determine the inventory value). Therefore it is not allowed to re-calculate the standard cost during the year.

Every year on January 1st, a calculation (cost roll) is run on every logistic company to update the standard cost of all items. The exchange rate used for the items purchased in foreign currency is the budget exchange rate defined in TLD Group procedure.

The standard cost cannot be modified during the year, except if the CFO decides to correct a mistake or anomaly in the initial standard calculation. The CFO is the only person who can decide to modify a standard cost. Other corrections during the year can be made under exceptional circumstances and with the prior Group CFO approval.

2. Spare Parts

Because it is critical for spare parts activity to have reliable MIP, the cost-roll rule is different for the SPH. Each month, the SPH manager shall provide the top 20 (at least) variances (positive & negative) analysis to the CFO / controller. Based on this analysis, the CFO / controller can decide to run a cost-roll on certain items.

3. Parts creation

For a new part number created in BaaN, the CFO / controller may, with the Regional CEO prior approval, delegate the cost-roll initiative to the Purchasing department. The ultimate responsibility still remains with Finance which must make monthly audits on the standard costs of the parts.

- Purchased items :

Per TLD rules, the calculation of the standard cost is taking the value in the BaaN field "*current price*" in the currency of the reference supplier, translated into the currency of the logistic company at the Budget rate. The *current price* is maintained by the purchasing department in the tab 3 of the item data and adds the standard allocation of freight in cost and of a cost of the logistics department (FIR : freight in rate + MBR : material burdened rate). The sum of FIR+MBR is commonly called *surcharge rate*. FIR and MBR are further defined below.

The *current price* is the reference price for the calculation of the standard cost. It is the most probable purchase price at the time of the cost-roll (before expected future increases) and is not necessarily equal to the Last Purchase Price. The *current price* is a price without transportation (FIR) or logistic (MBR) surcharge that will be applied separately, and before any slow moving depreciation.

Special cases:

- o In the case of a newly created item, the *current price* should not be the prototype price but the price for serial production¹.
- o In the case of prices linked to quantities or in the case of multiple sources, the current price of the item data (as well as the reference currency) must reflect the standard source of purchase and standard quantities, even though by exception the article may be purchased from a different supplier at a different price and in another currency.²

The *current price* will be updated on day-to-day basis by the Purchasing department based on the evolution of the purchase prices, *even though the standard cost will not be rolled until the following year*.

In all cases the purchasing department must make sure that before the end of the year and the general cost-roll on January 1st all the *current prices* in the system are updated. The future cost changes or cost reductions should not be anticipated before the general cost-roll. This way the PPV properly measures the variance between the purchasing costs of the current year with the purchasing costs of the previous year.

By set up, the cost-roll will use the following cascade to calculate the Standard cost: Current Price / Current Price / Current Price / Current Price. Therefore, if the Current price is not filled in, the item will result with a 0-standard cost and will be easily flagged for correction.

¹ There can be exceptions to this rule for very specific cases, such as DTV – to be approved by the Regional CFO

² In case there is only one FIR for the business unit (temporary situation until switch to double-FIR system), the standard source can reflect a mix of 2 sources, for instance 50/50 with different prices (in this case the current price will be the weighted average of the prices from the 2 sources)

- Manufactured items (sub-assemblies manufactured internally or sub-contracted)

The standard cost calculation uses the *current price* with surcharge of all components in the BOM, and adds the cost of labor which is the amount of labour hours defined in the routing multiplied by the standard burdened labor rate (BLR), and/or the cost of the subcontracted work.

When creating a manufactured item, the manufacturing department/planner must make sure that no item of the multi-level BOM has a zero standard price.

Parts with surface treatment (incl. painting) must have a distinct part number when not treated (PN #XXX – RAW).

It is reminded that no specific work order should be opened for sub-assemblies or sub-components dedicated to be assembled on units.

Responsibilities for data setup and maintenance

Several departments are involved or have an influence in the calculation of the standard cost, as they are responsible for the input and maintenance of the data in the ERP.

The responsibilities for the data setup, input and update are the following :

- « *current price* » - in tab 3 of item data : Purchasing department (sourcing engineer)
- BOM : Engineering department
- routing: Manufacturing department (during the year for the new projects, and end of the year - before the general cost-roll - on all project based on the budget assumptions).
- Standard rates (FIR, MBR, DLR, BLR) : Finance and cost accounting department (at end of year based on budget assumptions).

No item shall have a 0-Standard Cost. Therefore, accounting will run a monthly report showing all the active items which Standard Cost is 0. An active item is an item with a least a transaction date that is less than 1 year old or an item with a quantity on order. Any item appearing on this list shall be immediately corrected: the Purchasing Department will update the current price and then Finance will run the Cost Roll on this item.

Definition of FIR and MBR

- FIR

The FIR is equal to the sum of (i) custom & import duties, (ii) other taxes, and (iii) transportation costs to receive the parts, divided by the total parts purchase of the period

The objective is to have 2 FIR implemented for each company (one for regional purchases and one for long distance purchases).

Region EMEA entities switched to the 2-FIR system since 2013 January 1st. Region NALAAJ and Region CSEA entities will have to switch no later than January 1st 2014.

- MBR RAW

The Material Burden Rate (MBR) will be split into 2 parts, in order to value the raw material and WIP inventory with a different burden rate, as imposed by the IFRS accounting rules.

The MBR RAW covers the cost to put the parts into our inventory. It is equal to the sum of (i) warehouse and handling workers payroll pro rata the portion of their activity dedicated to bringing in the part in the warehouse, (ii) depreciation or rental of warehouse equipment (forklifts, shelves, IT...) pro rata the portion of their activity dedicated to bring in the part in the warehouse, divided by the Purchase value of the period.

- MBR WIP

The MBR WIP covers the costs to bring the parts from the raw material inventory into WIP. It is equal to the sum of (i) logistic and purchasing department costs, (ii) warehouse and handling workers payroll pro rata the portion of their activity dedicated to move the part out of raw material inventory and into WIP, (iii) depreciation or rental of warehouse equipment (forklifts, shelves, IT...) pro rata the portion of their activity dedicated to move the part out of raw material inventory and into WIP divided by the Purchase value of the period.

As of January 01, the rate applied is the coming year budget rate.

BaaN surcharge rate will be the sum of FIR + MBR RAW + MBR WIP and all the inventories will be value by BaaN based on this surcharge rate. Therefore, Accounting will have to pass a manual entry every month in order to withdraw the MBR WIP from the raw material inventory value.

Finance department will update the inventory value based on YTD actual FIR & MBR at the end of each quarter by a manual entry.

PPV

The Purchase Price Variance measures the difference between the Standard Cost (without surcharge) and the price invoiced by the vendor and reconciled. The PPV appears in 2 stages:

- upon receipt of the part in our inventory, the system registers the variance between the value of the PO in BaaN and the Standard Cost (without surcharge). It is the "PPV Standard"³;
- upon receiving the invoice from the vendor and approving it in BaaN, the system registers the variance between the value of the line of the PO and the value of the line of the invoice. It is the "PPV Invoice" .

PPV is obtained through BaaN and MIS department will issue within few weeks a routine to extract more easily the final result and some data to ease the analysis process.

To simplify the communication TLD will use internally the following definitions exclusively:

- **PPV STANDARD = PPV STANDARD NET + Forex impact on PO line**
- **PPV STANDARD NET = Standard Cost (PRS) W/O surcharge – Order Price expressed in the currency of TLD BU @ budgeted currency rate**
- **PPV INVOICE = PPV INVOICE NET + Forex impact on invoice**
PPV INVOICE NET = Order Price – Invoice Price expressed in the currency of TLD BU @ PO rate.

The PPV that we will monitor operationally will be PPV STANDARD NET and PPV INVOICE NET

³ The PPV Standard was previously called in certain BUs 'Expected PPV' or 'PPV Logistique' ; the PPV Invoice was previously called in certain BUs 'Unexpected PPV' or 'PPV Finance'

The PPV STANDARD NET will measure for each BU and each month the deviation between the Prices used for placing order to supplier and the price used for calculating the Standard Cost and it will be NET of all FX effects. In other words, this PPV will allow us to measure if we are placing PO to supplier at a price in line or not with our current Standard Costs

The PPV INVOICE NET will measure the for each BU and each month the deviation difference between the price invoiced by suppliers and the prices indicated by TLD in the PO to the supplier and it will be NET of all FX effects. In other words, this PPV INVOICE will allow us to control well if suppliers are respecting our PO'prices.

PPV Invoice Net must be kept as low as possible both in value and in quantity of lines since even a very small variance creates additional clerical work.

A "negative PPV" is a PPV unfavourable to TLD.

Please find a short synopsis showing the different impacts on the PPV.

